



OMNIKON

The Byte Rebels

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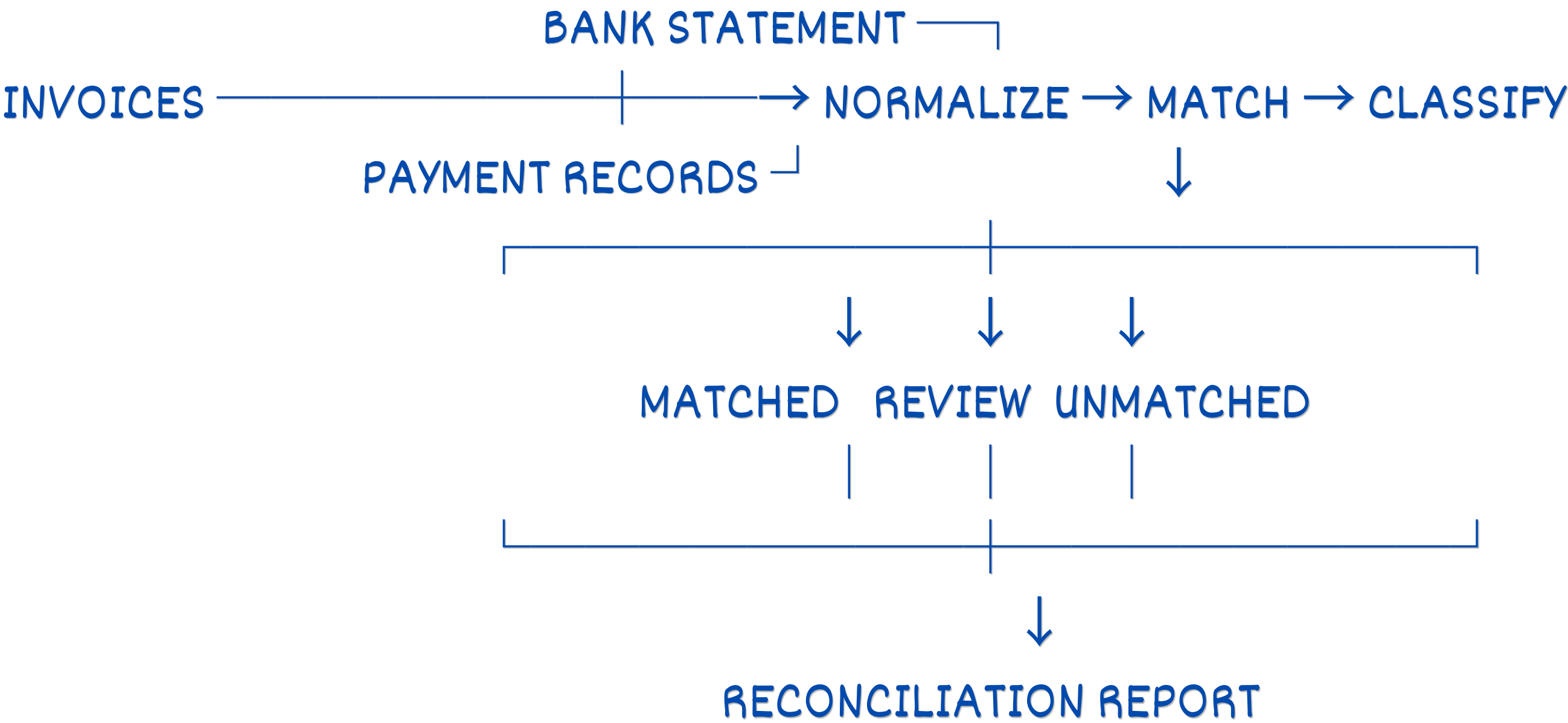
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FinTech

Automated Account Reconciliation for Small Businesses
Small businesses with limited accounting staff often spend excessive time reconciling accounts manually. Create a solution that automates and simplifies this process.



ReconcileX — Smart Account Reconciliation.



Automatically compares financial records, identifies discrepancies,
and prioritizes transactions requiring human review.

EXPLAINABLE MATCHING
Confidence Score + Matching
Evidence

Architecture & Implementation Flow

DATA INPUT

Bank Statements • Invoices • Accounting Records



DATA PROCESSING

CSV/PDF Parsing • Cleaning • Normalization



RECONCILIATION ENGINE

Amount • Date • Vendor • Reference Matching



AI ANALYSIS

Confidence Scoring • Duplicate • Missing • Anomaly Detection



DECISION & OUTPUT

Auto-Matched • Needs Review • Unmatched



DASHBOARD

Insights • Alerts • Approve / Reject / Edit

Tech Stack

Frontend

React.js • Tailwind CSS

Backend

Node.js • Express.js • TypeScript

Database

Supabase • PostgreSQL

AI & Intelligence

LLM • Rule-Based Matching • Confidence Scoring

Data Processing

CSV Parsing • PDF Text Extraction • Data Normalization

Deployment

Vercel (Frontend) • Render (Backend)

METHODOLOGY

01 — UPLOAD



02 — EXTRACT



03 — CLEAN



04 — MATCH



05 — DETECT



06 — REVIEW



07 — RECONCILE

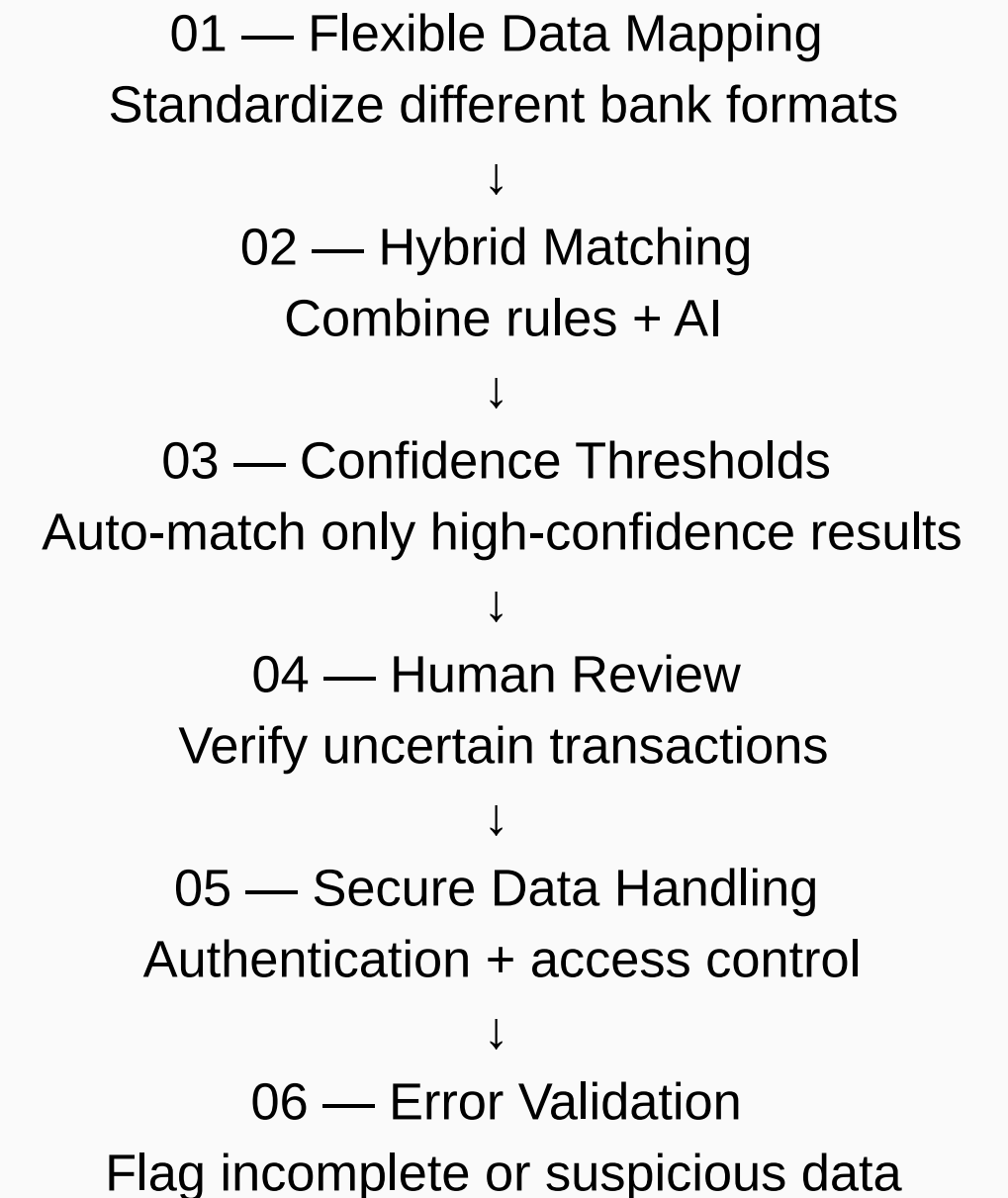
IDEA FEASIBILITY

- Technically Feasible — Uses proven web, database and AI technologies.
- Cost Effective — Can be prototyped using free/low-cost cloud services.
- Easy Data Integration — Supports CSV/PDF transaction data.
- Scalable — Can handle increasing transaction volumes.
- Practical & Useful — Reduces manual reconciliation effort.
- MVP Ready — Core features can be built and demonstrated

POTENTIAL CHALLENGES & RISKS

- Inconsistent Data: Different banks use different formats and transaction descriptions.
- False Matches: Similar transactions may be incorrectly matched.
- PDF Complexity: Scanned or poorly formatted PDFs may be difficult to extract.
- Data Security: Financial information requires secure handling.
- AI Errors: LLMs may misinterpret ambiguous transaction descriptions.

STRATEGIES TO OVERCOME



IMPACT AND BENEFITS

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TARGET AUDIENCE

- Small Business Owners
- Accountants & Bookkeepers
- Startups & Micro-Enterprises
- Businesses with Limited Accounting Staff

POTENTIAL IMPACT



SOCIAL

- Makes financial management easier for small businesses
- Supports teams with limited accounting resources



ECONOMIC

- Reduces accounting workload and operational costs
- Helps identify financial discrepancies faster



ENVIRONMENTAL

- Reduces paper-based documentation

OVERALL IMPACT



Less Manual Work
Automates repetitive tasks



Lower Cost
Reduces operational expenses



Fewer Errors
Better matching & accuracy



Faster Financial Insights
Faster reports & alerts



Less Work • Fewer Errors
Faster Decisions



RESEARCH AND REFERENCES

RESEARCH

- Sarumi et al. (2022) – Financial Accounts Reconciliation System Using Enhanced Mapping Algorithm.Account reconciliation becomes difficult due to heterogeneous financial data and increasing data volume. Manual and heuristic-based methods can become inefficient and prone to errors, increasing financial risk..The study proposes a computational data analytics model to improve reconciliation and support faster decision-making. Source Inderscience Publishers – Research Paper <https://www.inderscience.com/info/inarticle.php?artid=124843&utm>

- Munoz, Jalili & Tafakori (2025) – Enhancing Bookkeeper Decision Support Through Graph Representation Learning for Bank Reconciliation Bank reconciliation requires matching bank statement transactions with invoices and accounting records.Existing automated systems can struggle with complex one-to-many transaction matches. The research shows that data-driven approaches can improve matching and provide better decision support for reconciliation. Source:The Journal of Finance and Data Science – Research Paper <https://www.sciencedirect.com/science/article/pii/S2405918825000224?utm>

- Khan & Mita (2024) – Automated Financial Reconciliation Systems for Enhancing Efficiency and Transparency in Enterprise Accounting Workflows Manual matching and fragmented accounting systems can slow reconciliation and increase audit risk. The study found that automation capability, integration quality, and data readiness are important for improving workflow efficiency.Automated reconciliation can improve both efficiency and transparency. Source: Research Paper – Automated Financial Reconciliation Systems https://www.researchgate.net/publication/400471145_Automated_Financial_Reconciliation_Systems_for_Enhancing_Efficiency_and_Transparency_in_Enterprise_Accounting_Workflows?utm_s

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1. Sarumi et al. (2022) – Financial Accounts Reconciliation System –DOI 0.1504/IJBIDM.2022.124843 |
2. Muñoz et al. (2025) – AI-Based Bank Reconciliation – DOI: 10.1016/j.jfds.2025.100170 |
- 3.Khan & Mita (2024) – Automated Financial Reconciliation – DOI: 10.63125/0mf6qw97

Research Gap Identified

Most existing solutions are built for large enterprises or complex systems.

